

LAB 1 REPORT

STATISTICAL LEARNING & INTRODUCTION TO PYTHON (ISLR / ISLP CHAPTER 2)

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Course / Group: IT171IU – Statistical Learning (Lab Group 1)

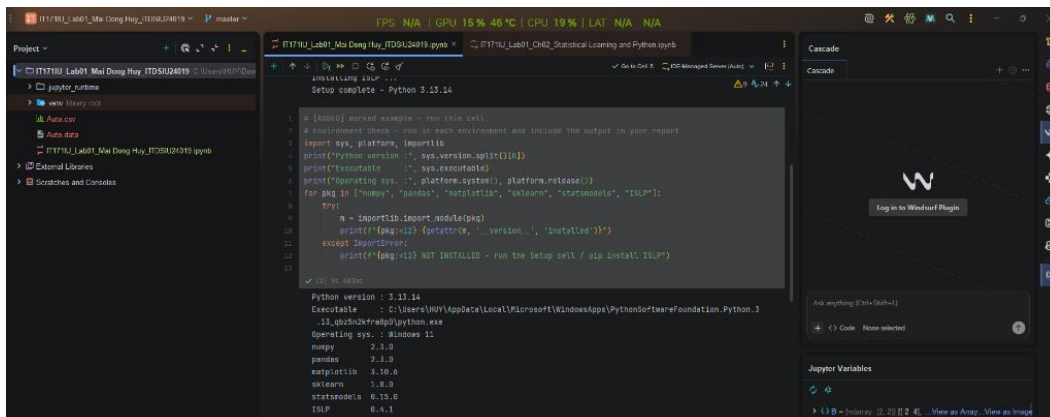
Instructor / Semester: Semester 1, Academic Year 2026

Task 0 – Environment Setup and Verification

The statistical learning Python environment was successfully set up and verified across both desktop IDE (Windows 11, Python 3.13) and Google Colab cloud environment.

```
Python version : 3.13.14
Executable      :
C:\Users\HUY\AppData\Local\Microsoft\WindowsApps\PythonSoftwareFoundation.Python.3.13_qbz5n2kfra8p0\python.exe
Operating sys.  : Windows 11
numpy           2.3.0
pandas          2.3.0
matplotlib      3.10.6
sklearn         1.8.0
statsmodels     0.15.0
ISLP            0.4.1
```

Check Message : Hello, Mai Dong Huy! My statistical-learning environment works.



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ISLP            0.4.1
```

Exercise 1 – NumPy Warm-up

Vector $v = (2, 4, \dots, 40)$ reshaped to 4x5 matrix M :

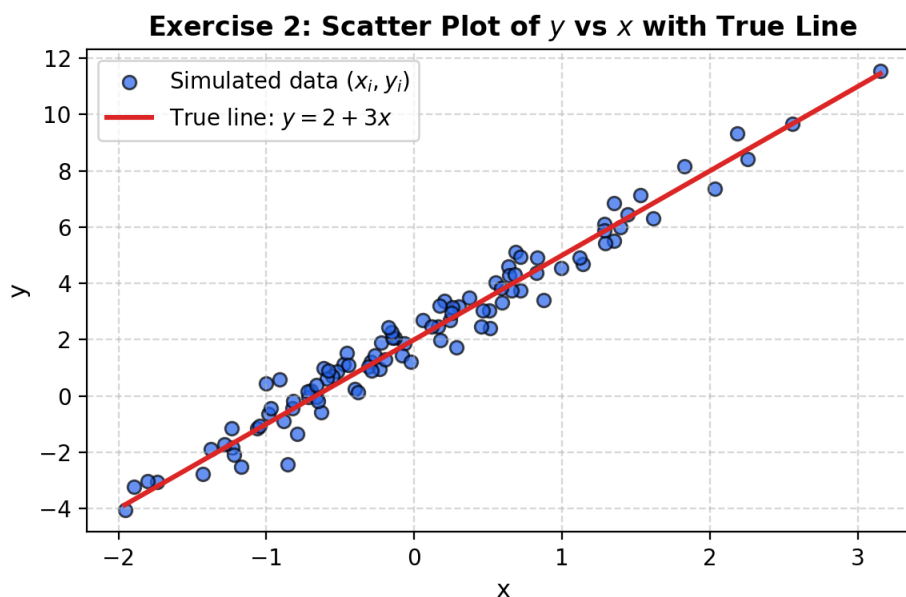
```
M.shape: (4, 5) | M.ndim: 2 | M.dtype: int64
Row means: [ 6. 16. 26. 36.]
Column std: [11.18, 11.18, 11.18, 11.18, 11.18]
View mutation test: B = M[:2, :2]; B[0, 0] = -1 -> M[0, 0] changed to -1 (True)
Copy mutation test: B_copy = M[:2, :2].copy(); B_copy[0, 0] = -1 -> M[0, 0] remained 2 (False)
```

Interpretation: *Slicing a NumPy array produces a view sharing memory with the parent array, causing modifications to B to alter M directly. In contrast, .copy() allocates distinct memory so mutations stay isolated.*

Exercise 2 – Random Numbers, Correlation & Saved Figure

Simulated $n = 100$ observations: $x \sim N(0, 1)$, $\text{eps} \sim N(0, 0.5^2)$, $y = 2 + 3x + \text{eps}$.

Empirical correlation between x and y : 0.9820



Interpretation: *The sample correlation (~ 0.9820) confirms an exceptionally strong linear relationship with minimal residual noise. The fixed random seed (2026) ensures strict reproducibility; removing the seed produces fresh pseudo-random numbers upon each execution.*

Exercise 3 – Cleaning the Auto Data

Loaded Auto.data with whitespace delimiter and '?' recognized as NaN.

```
Original shape: (397, 9) | Missing value rows: 5 | Cleaned shape: (392, 9)
Qualitative variables: origin (mapped to America, Europe, Japan), name
Quantitative variables: mpg, cylinders, displacement, horsepower, weight, acceleration, year
```

Quantitative Summary Table:

	range	mean	std
mpg	37.6	23.445918	7.805007

cylinders	5.0	5.471939	1.705783
displacement	387.0	194.411990	104.644004
horsepower	184.0	104.469388	38.491160
weight	3527.0	2977.584184	849.402560
acceleration	16.8	15.541327	2.758864
year	12.0	75.979592	3.683737

Interpretation: Exactly 5 observations contained missing horsepower records and were eliminated via `dropna()`. The remaining 392 observations show large differences in dispersion: weight and displacement have standard deviations of 849.4 and 104.6, whereas acceleration is much more tightly clustered.

Exercise 4 – Subsetting Rows with `iloc[]`

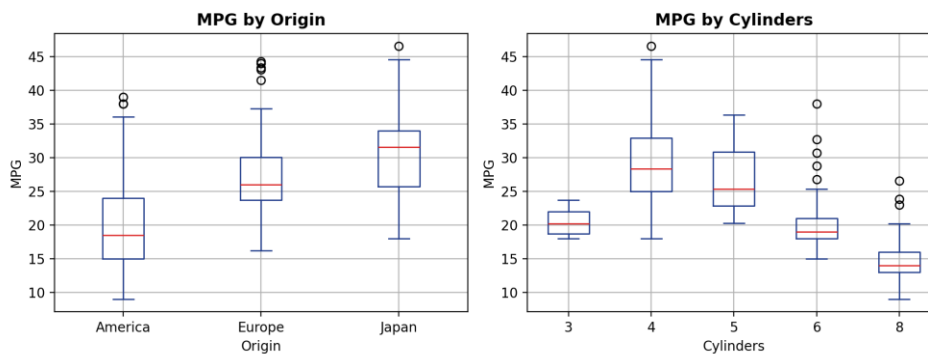
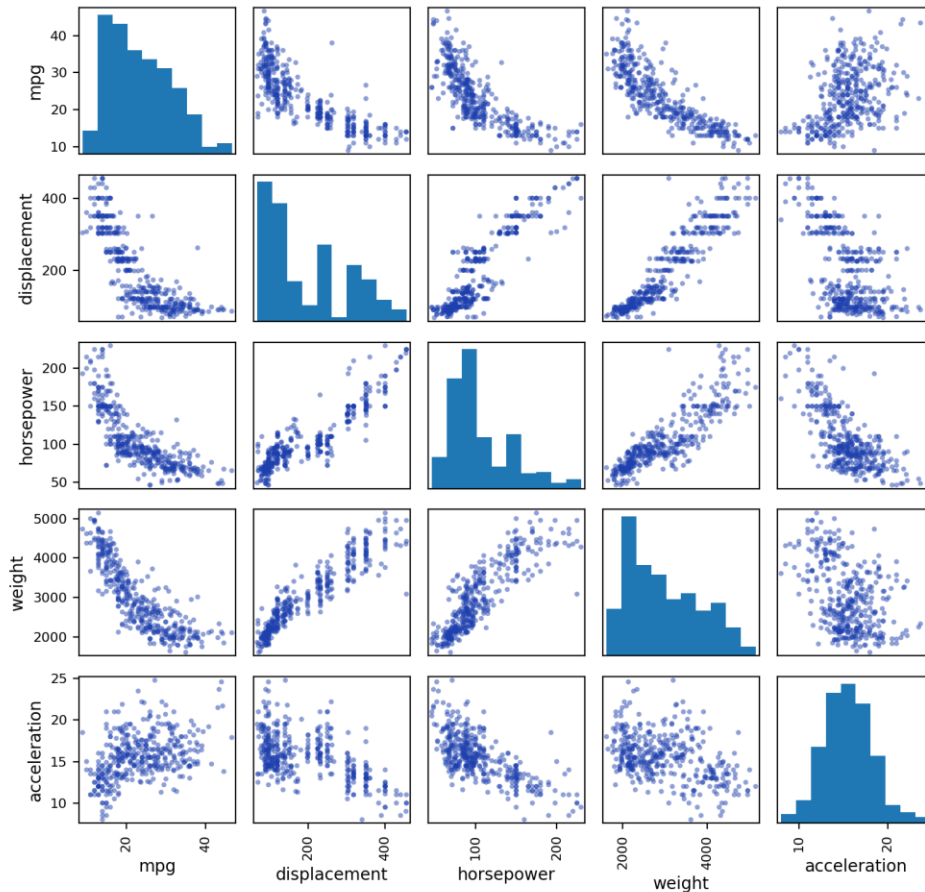
Removed observations 10 through 85 (1-based indices 9 to 84), retaining 316 rows:

Subset statistics comparison:

	Full Mean	Sub Mean	Delta Mean	Full SD	Sub SD	Delta SD
mpg	23.4459	24.4044	+0.9585	7.8050	7.8673	+0.0623
cylinders	5.4719	5.3734	-0.0985	1.7058	1.6542	-0.0516
displacement	194.4120	187.2405	-7.1715	104.6440	99.6784	-4.9656
horsepower	104.4694	100.7215	-3.7479	38.4912	35.7089	-2.7823
weight	2977.5842	2935.9715	-41.6127	849.4026	811.3002	-38.1024
acceleration	15.5413	15.7269	+0.1856	2.7589	2.6937	-0.0651
year	75.9796	77.1456	+1.1660	3.6837	3.1062	-0.5775

Interpretation: Weight and displacement experienced the largest drops (-41.61 lbs and -7.17 cu.in. in mean), while average MPG increased by +0.96. The removed rows were clustered in early-1970s heavy American automobiles; their removal systematically shifts the sample towards lighter, more fuel-efficient cars.

Exercise 5 – Graphical Exploration

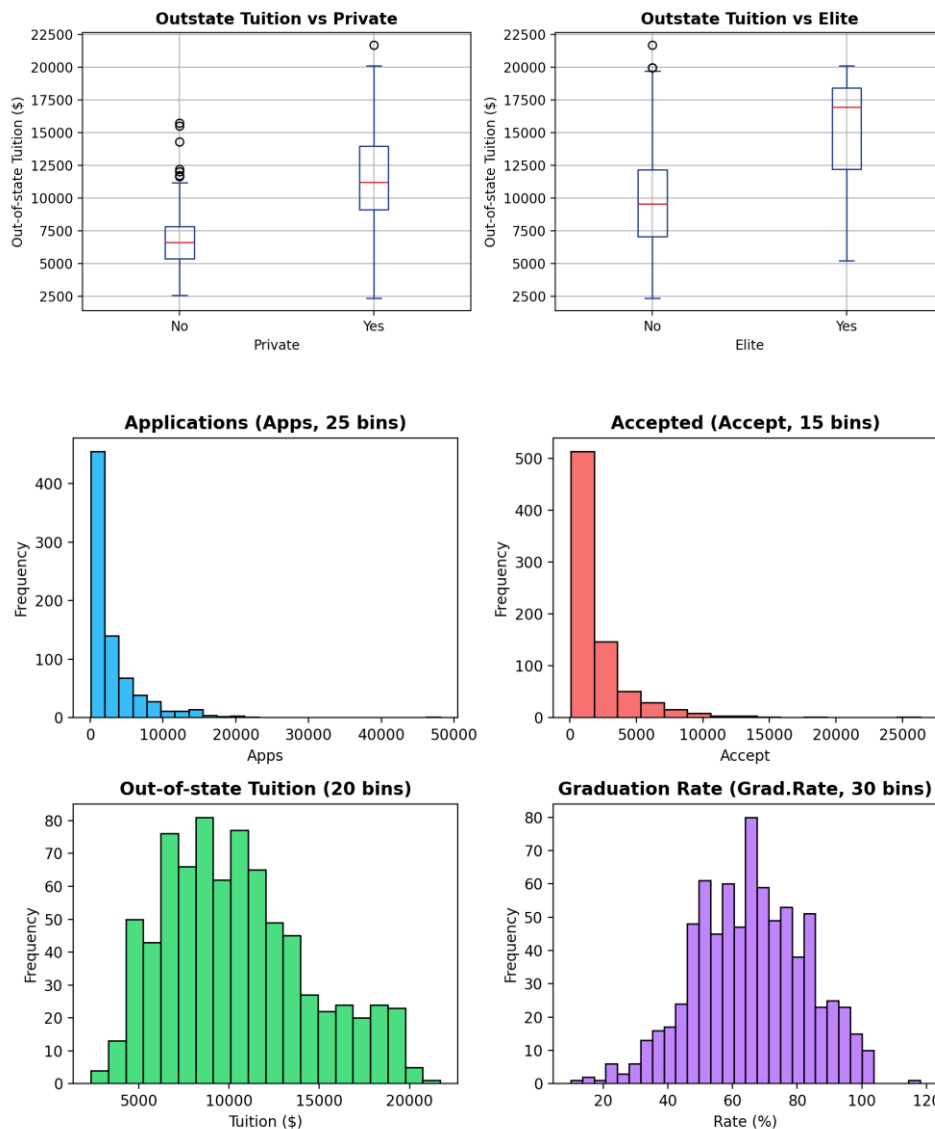


- Key Findings:** 1. *Non-linear relationship: Weight, displacement, and horsepower exhibit distinct convex non-linear negative relationships with mpg (diminishing fuel economy degradation for very heavy vehicles).*
2. *Cylinders effect: 4-cylinder vehicles achieve significantly higher median MPG than 6- and 8-cylinder vehicles.*
3. *Origin disparity: European and Japanese vehicles deliver noticeably higher fuel economy than American vehicles.*

Exercise 6 – The College Dataset

Exploration of 777 US Colleges from ISLP:

Total colleges: 777
 Private colleges: 565 (Yes) vs 212 (No)
 Elite colleges (Top10perc > 50): 78 (Yes) vs 699 (No)



Interpretation: Private and elite universities charge substantially higher out-of-state tuition than public and non-elite schools. Application (Apps) and acceptance (Accept) volumes show severe right-skewness, whereas tuition and graduation rates display approximately symmetric distributions.

Exercise 7 – Loops and String Formatting

Formatted loop iterating over quantitative columns:

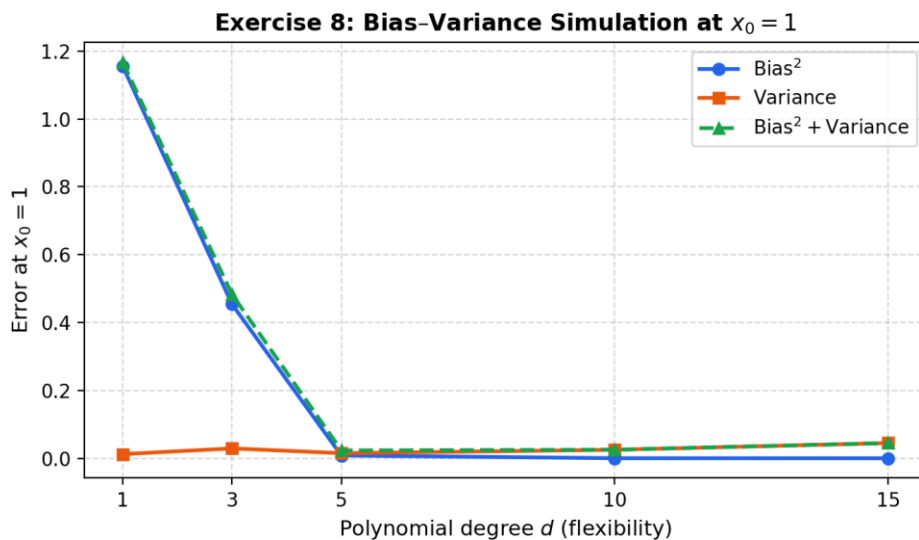
mpg	mean = 23.45	sd = 7.81	above-mean share = 47.4%
cylinders	mean = 5.47	sd = 1.71	above-mean share = 47.4%
displacement	mean = 194.41	sd = 104.64	above-mean share = 43.4%
horsepower	mean = 104.47	sd = 38.49	above-mean share = 37.8%
weight	mean = 2977.58	sd = 849.40	above-mean share = 43.4%
acceleration	mean = 15.54	sd = 2.76	above-mean share = 45.7%
year	mean = 75.98	sd = 3.68	above-mean share = 54.1%

Interpretation: Due to positive skewness in horsepower and displacement, relatively few vehicles with extreme engine sizes pull the mean upwards, resulting in less than 45% of cars exceeding the mean. In contrast, year exhibits 54.1% above-mean share.

Exercise 8 – Estimating Bias & Variance by Simulation (Challenge)

Simulation at $x_0 = 1.0$ with $n_{\text{train}} = 60$ over 200 replications:

d = 1:	Bias ² = 1.15608,	Variance = 0.01215,	Total MSE = 1.16823
d = 3:	Bias ² = 0.45406,	Variance = 0.02900,	Total MSE = 0.48306
d = 5:	Bias ² = 0.00817,	Variance = 0.01492,	Total MSE = 0.02310
d = 10:	Bias ² = 0.00016,	Variance = 0.02500,	Total MSE = 0.02517
d = 15:	Bias ² = 0.00000,	Variance = 0.04477,	Total MSE = 0.04477

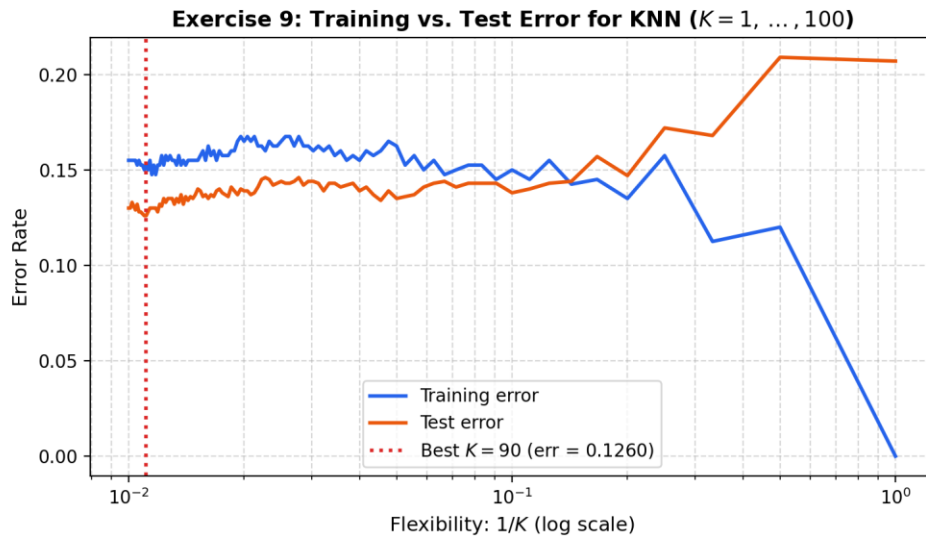


Interpretation: The empirical curves precisely corroborate statistical learning theory: as polynomial degree d increases, Bias² vanishes while model variance increases at high degrees due to noise overfitting. The total error curve forms a distinctive U-shape with the optimal trade-off achieved around degree $d = 5$.

Exercise 9 – Choosing K for KNN (Challenge)

Evaluating KNN classification test error over $K = 1, 2, \dots, 100$:

Optimal K: 90 (Minimum test error rate = 0.1260)
At $K = 1$: Training error = 0.0000, Test error = 0.2070
At $K = 90$: Training error = 0.1525, Test error = 0.1260



Interpretation: At $K = 1$ (highest flexibility, $1/K = 1$), training error is 0 but test error is high (0.2070) due to overfitting sample noise. Increasing K smooths the decision boundary, reducing test error to its minimum near $K = 90$ (0.1260). This faithfully reproduces Figure 2.17 in ISLR.